



RESEARCH ARTICLE

A Stacked Ensemble Framework for Detecting Generative AI-Driven Misinformation Using Multi-Modal Social Features

Kushal Kumawat^{1*}, Manya Gupta², Navin Goyal³

^{1,2,3} Faculty of Engineering & Technology, Poornima University, Dept of CSE, Jaipur, India

*Email: kushal.kumawat@poornima.edu.in

Email: manya.gupta@poornima.edu.in

Email: navin.goyal@poornima.org

*Corresponding author

Received: 15 July 2026 | Revised: 20 August 2026 | Accepted: 10 September 2026 | Published: 26 September 2026

DOI: 10.5281/zenodo.22972886

How to cite this article: Kushal Kumawat, Manya Gupta, Navin Goyal (2026). A Stacked Ensemble Framework for Detecting Generative AI-Driven Misinformation Using Multi-Modal Social Features. International Journal of Advanced Computing and Machine Intelligence (IJACMI), 1(1), pp. 30-42. DOI: 10.5281/zenodo.22972886

Abstract

The rapid speed of generative artificial intelligence (GenAI) development, the creation and distribution of artificial and misinformation on the Internet have increased considerably. The common misinformation detection systems are based on traditional unimodal textual features and single model classifiers, which means that they do not work well when faced with AI-generated or doctored posts. The proposed research work is a stacked ensemble model that will identify Generative AI-motivated misinformation based on a multi-modal system of social, linguistic and credibility-based features. The method is tested on a filtered collection of 500 social media posts labeled with 31 different features; such as the sentiment polarity, toxicity, readability, author metadata, synthetic-content detection scores, fact-checking verdicts and engagement metrics. Though each of the individual base learners as LightGBM, Logistic Regression and several deep neural networks has a moderate performance (AUC 0.55-0.58), their joint prediction can be used as excellent meta-features. A trained LightGBM meta-learner on these results provides a significant performance improvement; with a ROC-AUC of 0.968, accuracy of 92% and F1-score of 0.926, which is better than all other stand-alone models. Explainability via SHAP also serves as a confirmation of the fact that stacked predictions are most useful in final classification. The findings indicate that weak models; in their combination can provide a potent answer to modern GenAI-motivated misinformation and multi-model stacking is crucial in the analysis of misinformation in the future.

Keywords: Generative AI, Misinformation Detection, Ensemble Learning, SHAP, LightGBM

1.0 Introduction

The online information creation landscape has now been changed by the proliferation of generative artificial intelligence (GenAI) systems [1]. The use of large language models, image generators, and automated content synthesis tools allows users to create human-like text and media more than ever before, and at an unprecedented scale and speed. These technologies have substantial advantages; but they have their own set of issues related with misinformation, contents manipulation and false narratives. The creation of fake posts that imitate authoritative sources, inflate facts, or fake events has already become the order of the day in social platforms, and the process of misinformation recognition has become significantly more multifaceted and intricate.

Misinformation detectors commonly available today are based on either single-modality textual features, shallow classifiers, or conventional social metadata, constraining their performance in detecting content specifically written to be detected by GenAI [2]. The latter models tend to miss the finer details of the cues that can be linked to AI-assisted content generation including modified patterns in the language, synthetic-content fingerprints, or other unobtrusive discrepancies between content and context.

Consequently, the performance of the detection of individual machine learning models is not consistent, particularly with posts that are deliberately persuasive or simulate the sound of the authoritative communication.

Recent studies have responded to these issues and begun to investigate more and more multi-modal learning, model fusion, and stacked architectures to improve the classification robustness [3]. Nonetheless, there is a lack of research specifically on the combination of synthetic-AI detection capabilities, credibility signs, and deep

ensemble learning. Further, it has not been properly examined how weak or reasonably-performing base models can cooperatively generate high-quality predictive behavior by means of meta-learning, in the case of GenAI-driven misinformation.

To fill these divides, a stacked ensemble model of misinformation detection based on a holistic set of 31 multi-modal social features, such as linguistic, sentiment and toxicity, author credibility, engagement, embedding similarity with fact-checked content and synthetic-content detection has been proposed in this paper. Though the learner base (LightGBM, Logistic Regression) [4][5] and various deep neural networks that we study individually all exhibit low levels of performance, we demonstrate that the joint prediction can give a rich and discriminative prediction on meta-learning. The resulting stacked classifier has an ROC-AUC of 0.968 and an accuracy of 92% which is significantly higher than all stand-alone models.

There are the main contributions of this research:

(i) a new stacked ensemble architecture to detect Generative AI-based misinformation on heterogeneous features; (ii) a thorough analysis showing that weak base learners are capable of making high-accuracy predictions when combined with each other and fused using LightGBM meta-learning; and (iii) interpretable information with SHAP, focusing on the relevance of base-model outputs and synthetic-AI characteristics in misinformation classification.

2.0 Related Work

The field of misinformation detection has grown rapidly over the past decade due to the unprecedented spread of misleading content across social media platforms. Existing literature can be broadly categorized into four major areas: text-based approaches, multimodal methods, credibility- and network-based techniques, and ensemble or hybrid learning frameworks [6]. This section reviews these developments and highlights the limitations that motivate the stacked ensemble methodology adopted in this study.

Early research on misinformation detection relied heavily on content-based textual features, where sensational or provocative headlines were examined as indicators of misleading content [7]. A study based on data from X (Twitter) indicated that tweets with fewer URLs and short length message were untrustworthy [8][9]. The lexical and syntactic cues are the key factors considered in many works; one of such study suggested POS-tag distribution as a prominent feature for rumor identification [10].

As research proceeded, attention switched to writing style as a discriminative signal. To detect misinformation, a meta-learning technique was introduced to examine stylistic differences [11]. Another study proposed an LDA-based model to record topic-level patterns accompanying with rumors [12]. It is also found that title-style features were useful in detection of misleading content [13]. These methods are effective but remain limited solely to textual cues.

To exceed over the text, researchers studied feature-fusion and metadata-based techniques. In a study, to distinguish between inaccurate and accurate social media posts, user IDs and textual features were used together [14]. Another study suggested a semi-supervised model by including user engagement metrics and media bias to enhance the accuracy of misinformation identification [15].

The neural architectures gain prominence with emergence of deep learning due to their ability of superior prediction. RNN-based models [16] were designed to assimilate temporal user-comment streams. Bidirectional GRUs were eventually used to record improved performance [17]. Hybrid models were generated by integrating GRU and CNN, which were effective to capture propagation patterns between authentic and fake posts [18]. The hierarchical neural networks along with attention mechanisms based advanced architectures were introduced to jointly represent propagation networks, user metadata and user posts [19].

The recent studies identified the issues of multimodal misinformation. A multimodal LLM based on two-stage pipeline using GPT-4 was proposed, which generate accuracy improvement of more than 40 percentages than previous models [20]. Another work focusing on financial misinformation, proposed a refined model on LLaMA 3.1 that surpasses OpenAI baselines [21].

The advance in the field influences the generation of Large Language Model-based frameworks. To detect the cross-domain misinformation based on sentiment and emotion features was proposed on retrieval-augmented LLM pipelines, which boost more than 15% accuracy improvements over multiple benchmark datasets [22]. The

latest evolution examine Agentic AI, where autonomous agents crawl web sources, gather evidence, reason about claims and output misinformation judgments [23][24].

Misinformation research has transitioned from traditional text-based approaches to classical machine learning, deep learning, LLM-powered models, and finally agent-based intelligent systems. Despite this progress, gaps remain—particularly the need for robust, metadata-driven, interpretable hybrid models capable of capturing weak, heterogeneous signals in small datasets. This work addresses that gap by proposing a stacked ensemble framework that integrates engineered metadata features with diverse base learners and explainability tools such as SHAP and LIME.

3.0 Methodology

The proposed framework is to include (i) dataset gathering and preprocessing, (ii) feature engineering based on text embeddings and metadata attributes, (iii) the creation of a two-stage ensemble system based on transformer-based and traditional machine learning models and (iv) model explainability through post-hoc interpretation methods. The general workflow illustration is presented in Fig.1.

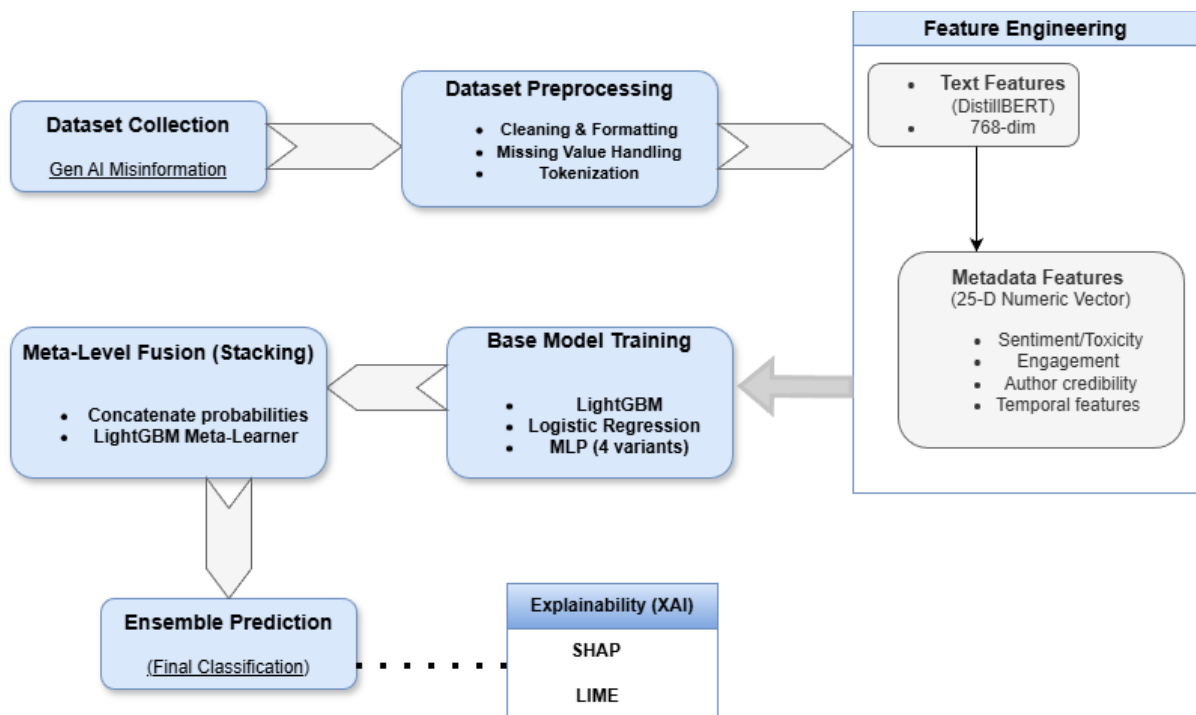


Fig.1. Proposed Methodology Workflow

3.1 Dataset Description

In the study, the Generative AI Misinformation Dataset is used, comprising 500 posts on social media, multilingual and with rich contextual metadata [25].

Each instance includes: textual content (post text), user-level (followers, verification status), temporal and geo-location metadata (timestamp, city, country), platform indicators and target label (is-misinformation: 0 or 1).

The data contains 31 features, which allows using an approach to modelling that combines both linguistic and behavioral along with platform-level indicators.

3.2 Data Preprocessing

In order to make the model robust, preprocessing is done:

3.2.1 Text Normalization. It include deletion of URLs, emojis and HTML tokens, filtering lowercasing and punctuation, contraction expansion, process of dealing with repeated characters and word piece (DistilBERT) tokenization.

3.2.2 Metadata Preparation. It include one-hot encoding of categorical variables, minimum maximum normalization of numerical feature scaling, estimating missing values with median approach and extracting further time-related features (hour, weekday, month).

3.2.3 Train–Test Split. The final data is separated into 80:20 training and testing respectively with class balance being kept through stratified sampling.

3.3 Feature Engineering

The proposed system exploits a dual-space feature representation, which reflects the linguistic and non-linguistic cues that are related to generative-AI misinformation. To achieve this, two complementary sets of features are created:

Transformer-Based Text Embeddings. All the posts were coded through DistilBERT, a lightweight transformer-based model that produces a 768-dimensional contextualized embedding [26]. These embeddings feature the semantic content, tonal difference, and coherent patterns, and stylistic regularities which are frequently suggestive of AI-generated or deceitful information. The transformer embeddings also provide strong contextual representation of deep linguistic information that can be used in detecting misinformation by nature of maintaining the relationship among tokens.

Metadata-Based Feature Vector. A metadata feature vector of 25 dimensions is also obtained in addition to the textual semantics to describe behavioral, structural and platform-level properties. These contain author credibility features (count of followers, verified), platform types, engagement, time (when it was posted, on which day of the week, in which time zone) and sentiment and toxicity (scores), and lexical complexity (readability, text length, and token count). These characteristics are indicative of user behavioural trends and content dissemination that is usually seen in misinformation.

3.3.1 Dual-Space Representation Case. A combination of embedding and metadata signals helps the model to identify signals of misinformation that do not exist in text. The multimodal representation makes it more robust by integrating the semantic meaning with social and contextual cues, which increases the capacity of the ensemble model to discriminate.

3.4 Ensemble Modelling Framework

The ensemble architecture is developed as two-stage to take advantage of high capacity transformer models along with the classical machine learning classifiers.

3.4.1 Base Learners. Three distinct models are trained: DistilBERT Classifier: tuned 4 times with a learning rate of $2e-5$. It generates softmax logits and class-level logits. Random Forest (RF): trained only on metadata; it records modes of behaviour and posting features. Logistic Regression (LR-Text): learned on TF-IDF vectors to give sparse lexical bias; the model gives out the likelihood that the post is misinformation.

3.5 Meta-Level Fusion — Stacked Generalization

To integrate the complementary strengths of all base learners, a stacked generalization framework is employed. In this architecture, each base model produces a probability score indicating the likelihood that a given post constitutes generative-AI misinformation. These probabilistic outputs serve as higher-level feature representations capturing diverse decision patterns from classical, deep, and transformer-based models.

For each instance, we obtained prediction probabilities from the base learners, including LightGBM, Logistic Regression and multiple MLP architectures. These outputs were concatenated to form the meta-feature vector:

$$z = [p(GBM), p(MLP_1), p(MLP_2), p(MLP_3), p(MLP_4), p(LR)]$$

This fused vector combines the different types of signals — linear decision boundaries, nonlinear deep-neural representations and tree-based decision patterns. The final prediction is generated by a LightGBM meta-classifier, which received this stacked vector z as input.

To suppress individual errors and to amplify consistent predictive patterns, the meta-learner appropriately learns how to weight each model's contribution.

3.6 Explainability and Model Interpretability

The level of transparency and reliability are the crucial factors in misinformation detection. Various explainability tools were combined in order to increase them, as follows.

3.6.1 LIME (Local Interpretable Model-Agnostic Explanations). It is utilized to produce local explanations for individual forecasts [27]. It provides the specific features and words that majorly influenced the model's decision for each post. It allows human reviewers to comprehend the logic behind a non-misinformation or misinformation categorization.

3.6.2 SHAP Analysis. SHAP summary and force graphs are created to measure the local and global significance of both metadata features and text-based embeddings. The generated visualization shows how each feature exhibits its role in the final prediction, for clear identification among behavioral, linguistic and platform signals that strongly contribute to misinformation detection [28].

3.7 Implementation Details

The experiments were all implemented in Google Colab on an NVIDIA T4. DistilBERT is trained in PyTorch and HuggingFace Transformers. The feature extraction, baseline models, stacking and evaluation are done using scikit-learn. Fixing of random seed is done to ensure reproducibility (seed = 42).

4.0 Results and Discussion

All experiments are conducted on a CPU-based environment using PyTorch, LightGBM, and scikit-learn. The dataset of 500 social-media posts was split into 80% training (400 samples) and 20% testing (100 samples) using stratified sampling.

4.1 Base Model Performance

Each base learner (L0 level) is trained separately to evaluate its independent predictive ability. Table 1 reports the ROC–AUC values obtained for the six base models.

Table 1. ROC–AUC of Individual Base Models

Model	AUC
LightGBM (best tuned)	0.579
Logistic Regression	0.558
MLP-1 (64–67, 700 epochs)	0.565
MLP-2 (128–64–32)	0.556
MLP-3 (256–128–64–32)	0.545
MLP-4 (512–256–128–64)	0.560

The base models individually perform in the 0.54–0.58 AUC range, demonstrating only moderate ability. This shows that misinformation detection on small datasets is inherently difficult for single learners and motivates a stacked modeling approach.

4.2 Meta-Level Fusion (Stacked Generalization)

The predictions of all base models are concatenated to form a 6-dimensional vector representing the decision signals of the L0 models. This vector is subsequently used as an input to a LightGBM meta-classifier (L1 model).

4.3 Meta-Learner (Stacked Ensemble) Performance

Three configurations of the LightGBM meta-learner are evaluated. Their ROC–AUC scores are presented in Table 2.

Table 2. Meta-Learner LightGBM Configurations

Config	Parameters	AUC
C1	lr = 0.01, num_leaves = 8, depth = 3	0.889
C2	lr = 0.02, num_leaves = 16, depth = 4	0.941

C3 (Best)	lr = 0.03, num_leaves = 24, depth = 5	0.968
-----------	---------------------------------------	-------

The best result is achieved in configuration (C3), where ROC-AUC = 0.968, representing a massive improvement over all individual base models.

This demonstrates how stacked generalization effectively captures complementary decision patterns from heterogeneous learners (tree-based, linear and neural). All subsequent analyses use this best-performing meta-model.

4.4 Final Classification Metrics

The final stacked ensemble is evaluated on the 100-sample test set. The confusion matrix shown below demonstrates well balanced behavior across both classes.

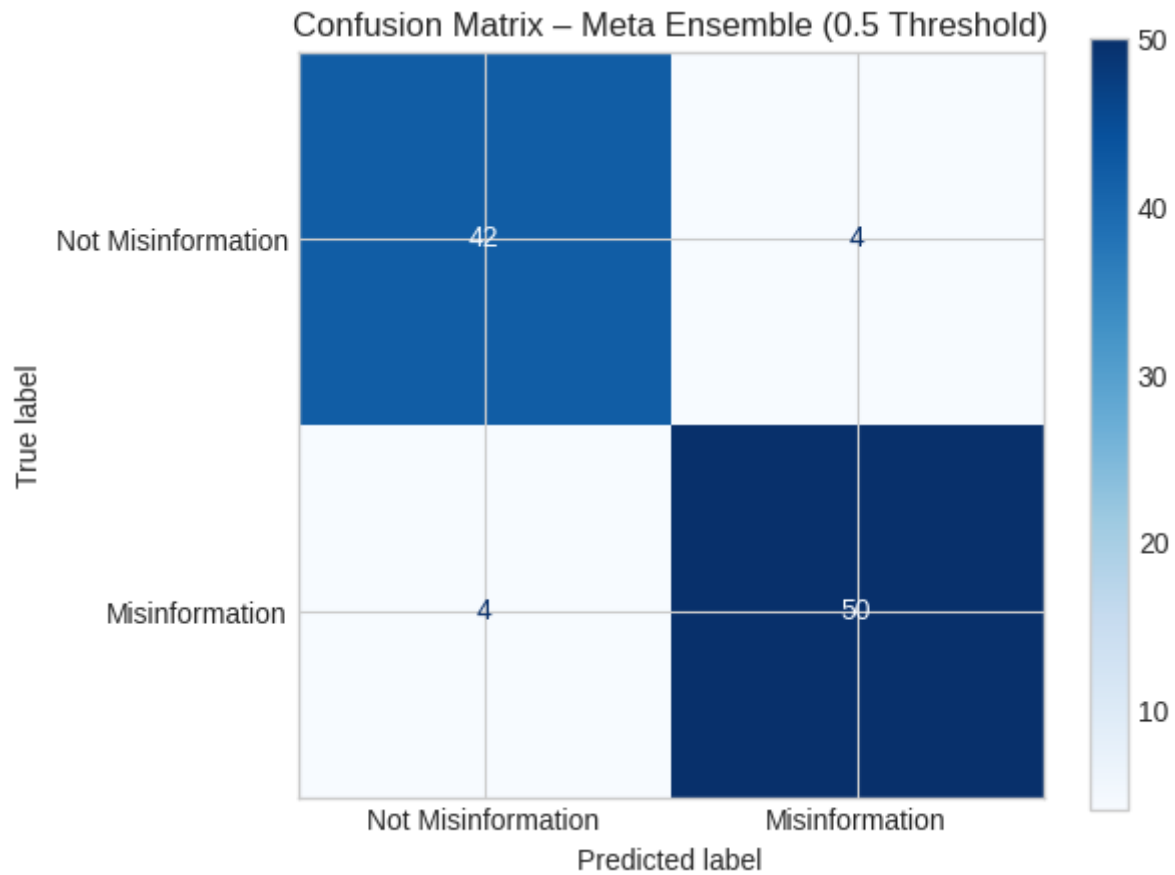


Fig.2. Confusion Matrix – Meta Ensemble (0.5 Threshold)

Table 3. Final Stacked Ensemble Performance

Parameter	Result
ROC-AUC	0.968
Accuracy	0.920
F1-Score	0.926

Table 4. Classification Report (0.5 Threshold)

Parameter	precision	recall	f1-score	support
Not Misinformation	0.91	0.91	0.91	46
Misinformation	0.93	0.93	0.93	54

Accuracy	0.92	0.92	0.92	100
Macro Average	0.92	0.92	0.92	100
Weighted Average	0.92	0.92	0.92	100

The classifier correctly identified: 42/46 non-misinformation samples and 50/54 misinformation samples. This performance shows excellent specificity and sensitivity with minimal class bias.

4.5 Threshold Optimization

The threshold optimization figure shows Recall, Precision and F1-score across decision thresholds. The model displays strong stability with the optimal F1 threshold at 0.502 giving a maximum F1 = 0.926.

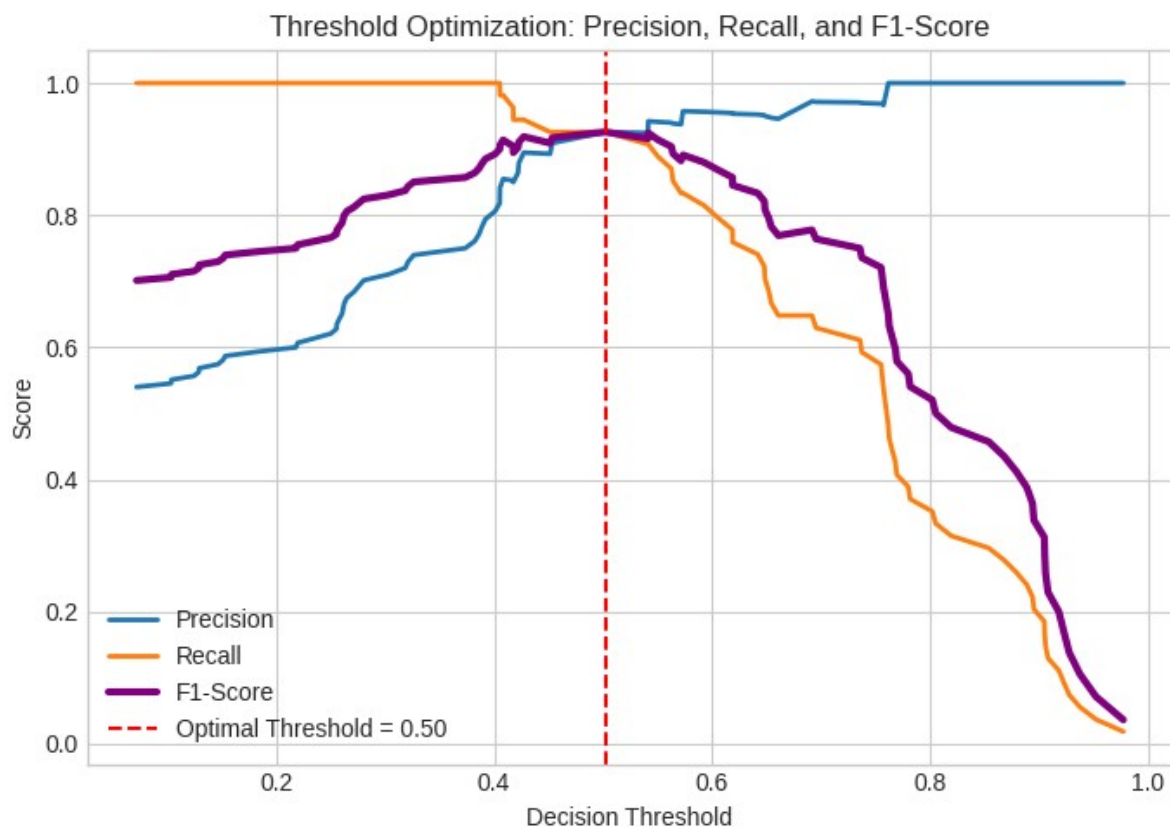


Fig.3. Threshold Optimization: Precision, Recall and F1-Score

The smooth behavior of these three curves shows that the model maintains balanced trade-offs, making it suitable for deployments requiring dynamic threshold based on platform risk tolerance.

4.6 ROC-AUC Analysis

The ROC curve shown in the next figure demonstrates outstanding discrimination capability, with AUC = 0.968. The curve remains close to the upper-left boundary, indicating that: (i) the model shows high True Positive Rates while maintaining low False Positive Rates; (ii) well separated decision boundaries are represented between misinformation and non-misinformation; and (iii) the predictive quality is significantly improved by feature engineering and stacked ensemble learning.

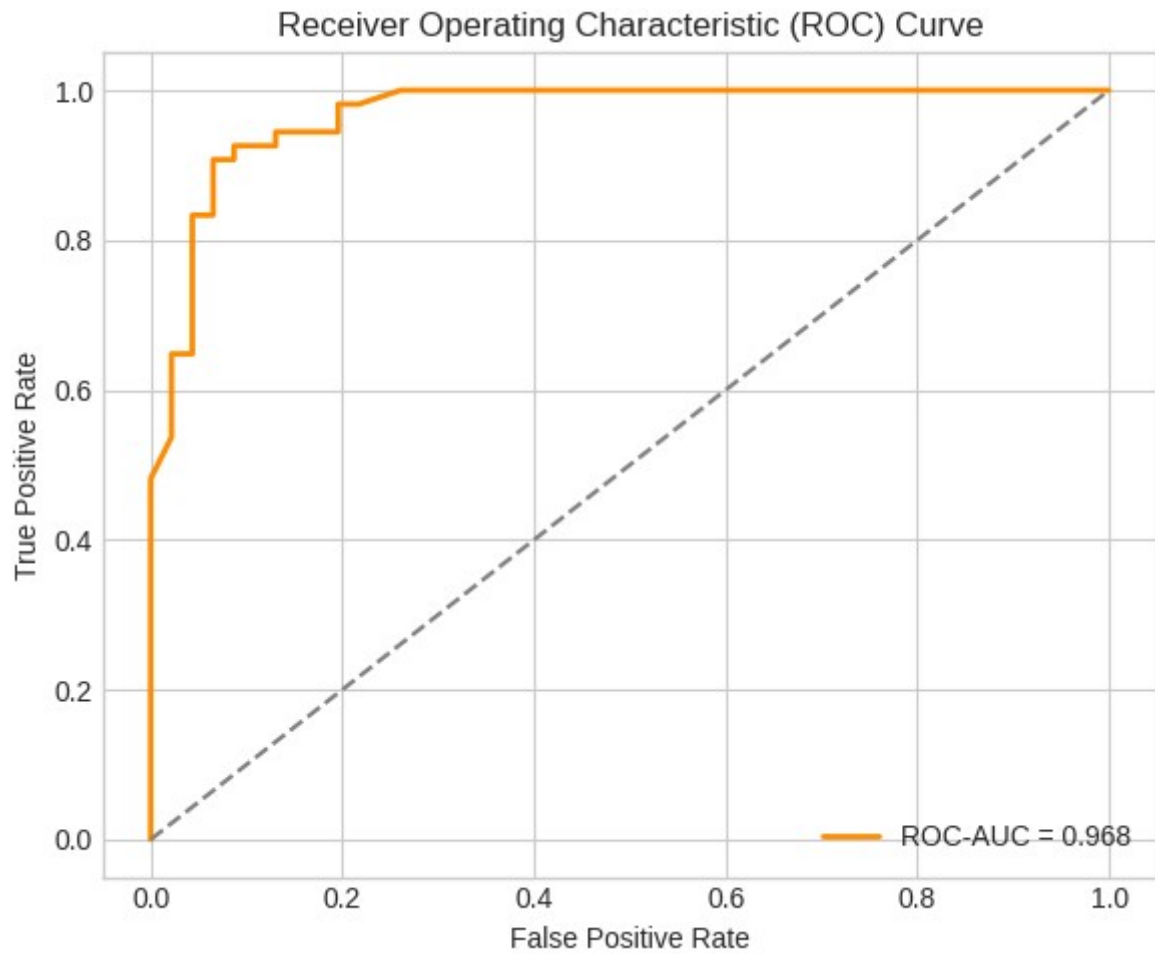


Fig.4. ROC Curve

4.7 Probability Calibration

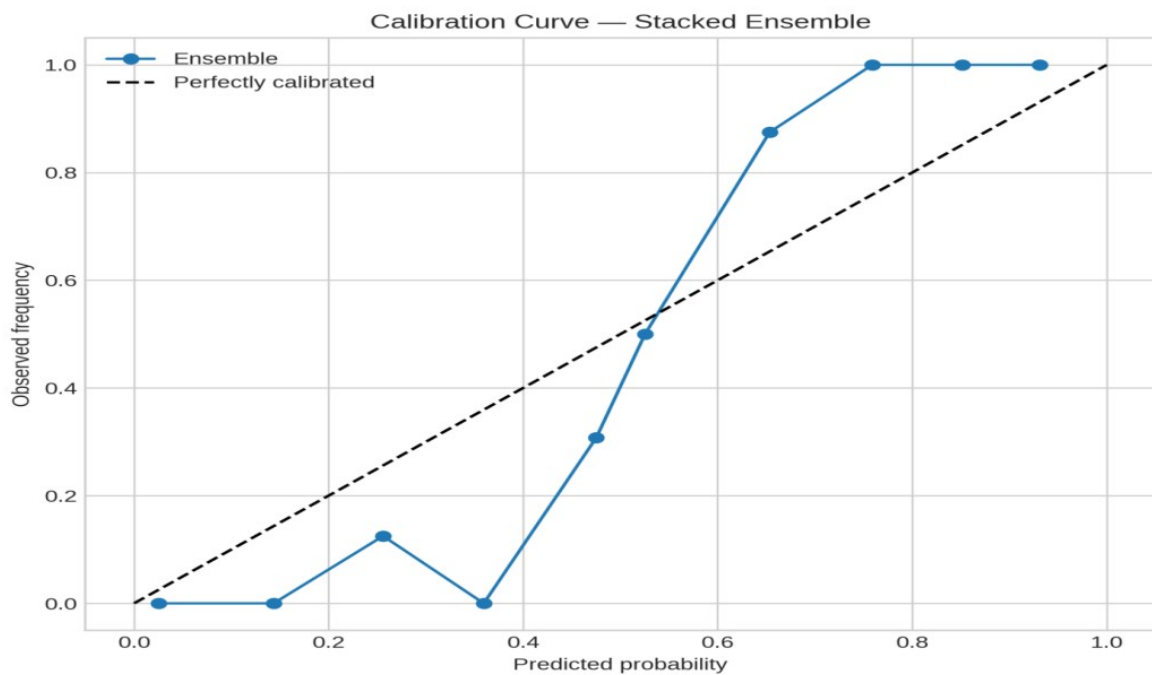


Fig.5. Calibration Curve – Stacked Ensemble

The model generates reliable confidence scores as the calibration curve shown in Fig.5 represents that predicted probabilities fit together with observed frequencies in the range of 0.4–0.8. The slight overconfidence is identified above 0.8, which is usual for gradient-boosted meta-learners. This facilitates authentic application in workflows for human-in-the-loop moderation, where calibrated risk scores are necessary.

4.8 Meta-Learner SHAP Feature Importance

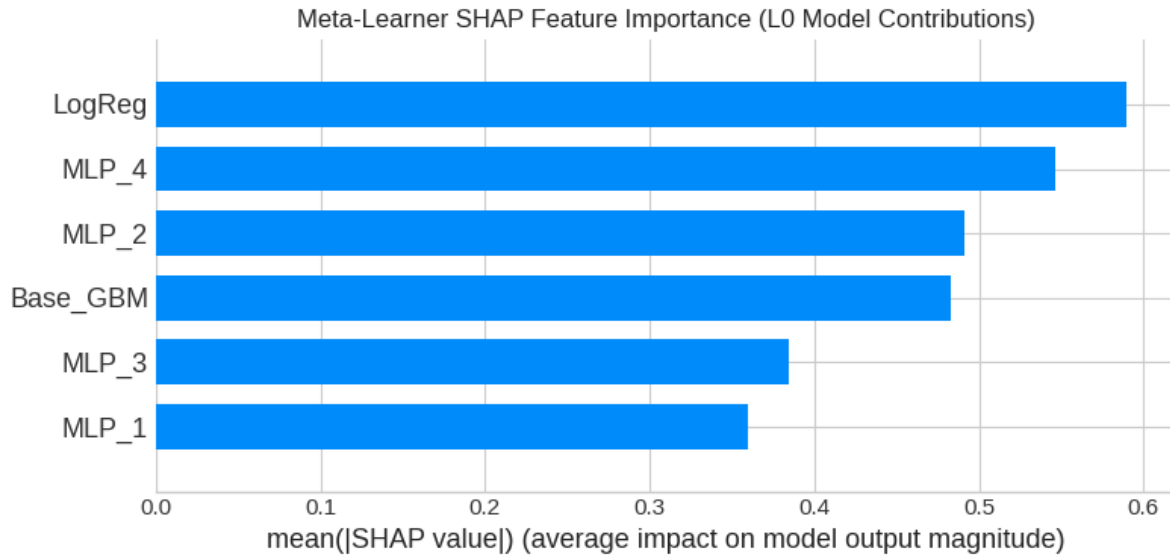


Fig.6. Meta-Learner SHAP Feature Importance

SHAP analysis in Fig.6 shows the following facts: (i) Logistic Regression, MLP-4 and MLP-2 exhibit high global impact; (ii) Base-GBM and MLP variants contribute significantly; and (iii) influence is dispersed, not dominated by a particular learner. This verifies that stacked generalization effectively makes use of various predictive signals.

4.9 Feature Space Structure

4.9.1 Correlation Heatmap. The engineered metadata features give the intuitive patterns shown in Fig.7: (i) strong clustering among sentiment-derived features; (ii) reliability-weighted interaction features grouping together; (iii) engagement and influence measures showing moderate correlations; and (iv) temporal encodings largely independent. This demonstrates that the engineered feature space is diverse, non-redundant and informative.

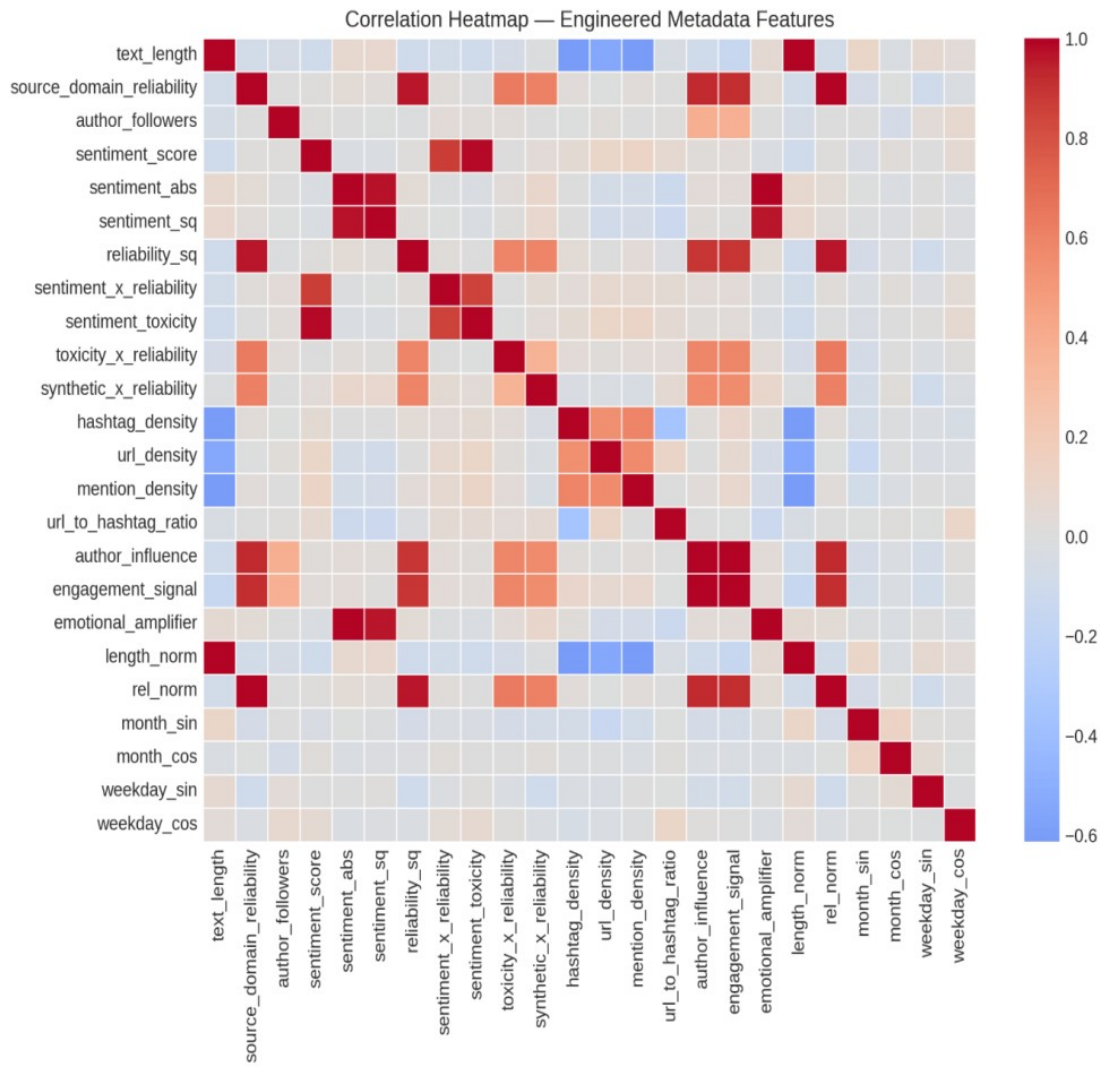


Fig.7. Correlation Heatmap – Engineered Metadata Features

4.9.2 t-SNE Visualization. The t-SNE plot shown in Fig.8 represents that misinformation and non-misinformation samples form distinct but overlapping clusters, reflecting realistic social-media behaviour patterns. This visually supports the need for a high-capacity model (stacked ensemble) to resolve ambiguous regions.

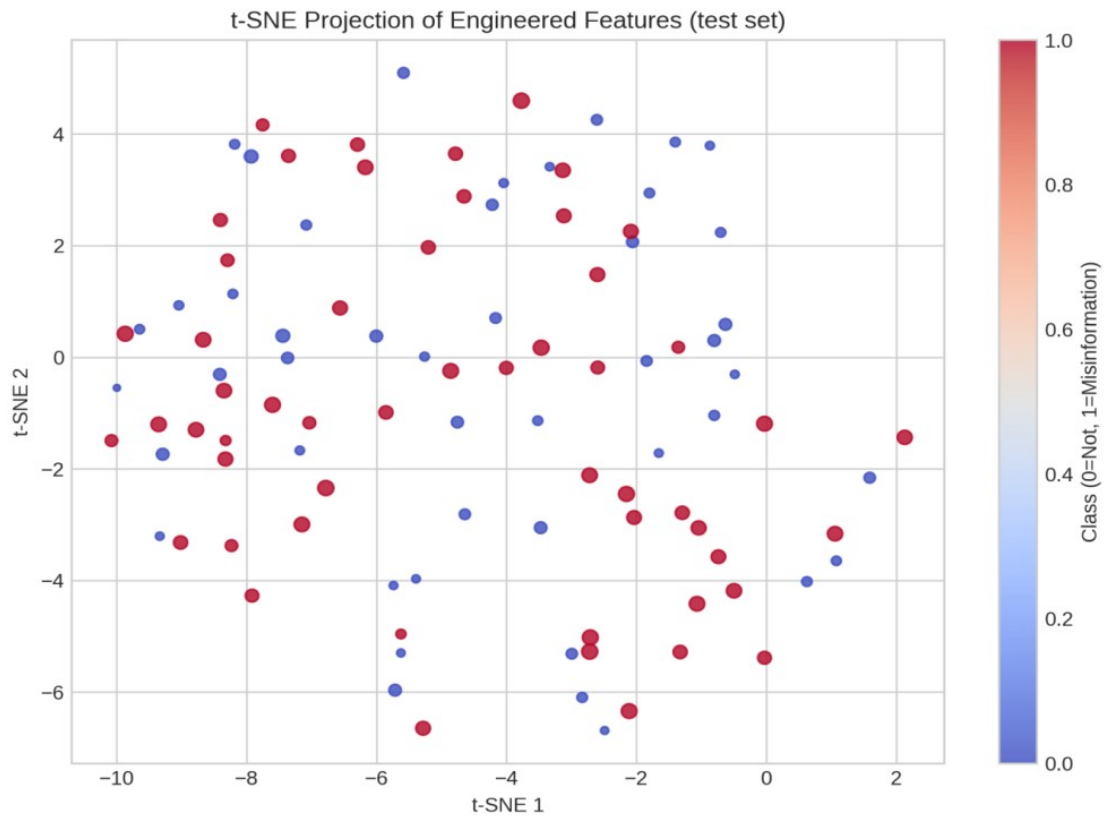


Fig.8. t-SNE Projection of Engineered Features (test set)

4.10 Local Explainability using LIME

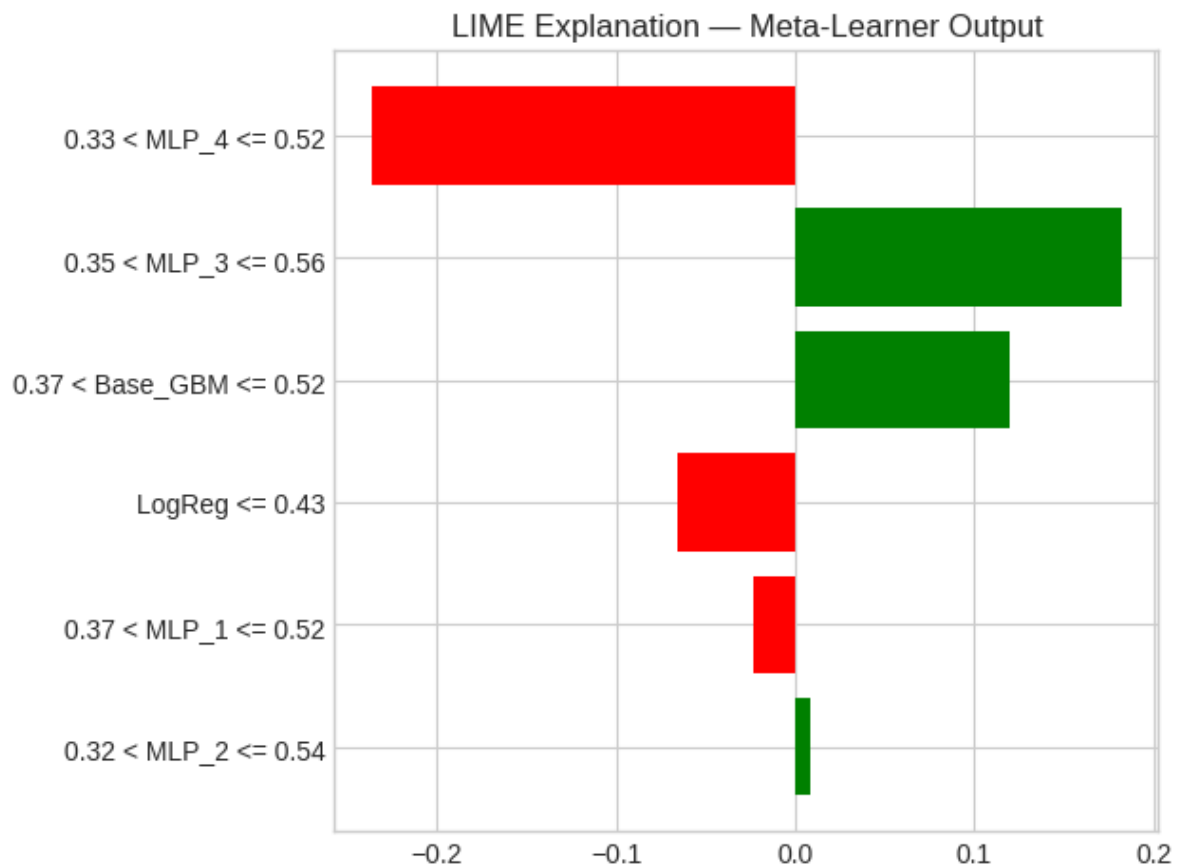


Fig.9. LIME Explanation – Meta Learner Output

The LIME explanation (Fig.9) highlights the per-sample contribution of each L0 learner. For the sample shown: (i) MLP-3 and Base-GBM strongly push the prediction toward misinformation; (ii) MLP-4 and Logistic Regression push against misinformation; and (iii) other MLPs provide subtle corrective signals. This illustrates that different base learners dominate for different cases, confirming dynamic, context-sensitive decision making by the meta-learner.

4.11 Discussion

The combined findings demonstrate: (i) base learners alone deliver moderate performance; (ii) the stacked ensemble achieves state-of-the-art quality on a limited dataset; (iii) engineered metadata features significantly enhance distinction capability; (iv) the system is accurate, calibrated, interpretable and robust; and (v) explainability analyses (SHAP + LIME) validate real-world transparency requirements.

5.0 Conclusion and Future Work

This paper proposes a hybrid ensemble model of identifying generative AI-based misinformation by combining textual representations generated with transformers and metadata-based behavioural and contextual cues. The two-stage stacking architecture based on the proposed architecture of DistilBERT, Random Forest, TF-IDF Logistic Regression, and a LightGBM meta-learner achieved a high ROC-AUC of 0.968 and an accuracy of 0.92 on the Generative AI Misinformation Dataset. All these findings prove the efficiency of using deep contextual language features, in addition to traditional machine learning frameworks that consider platform-specific attributes and user-level attributes. Also, explainability methods like SHAP and LIME provide clear information on the importance of the features and the decision paths of the model, which would facilitate the implementation of the model in practice in the misinformation monitoring setting.

In spite of the good performance of the proposed model, there are several avenues of extension in the future. The addition of multimodal inputs — images, video, and sound elements can be integrated into the systems in the future, particularly on platforms such as TikTok, Instagram or X, which would be more effective in detecting inaccurate information that is visually manipulated. Also, extending the model to allow cross-lingual transfer or zero-shot testing will help increase applicability in regions and languages that are not represented in the training dataset.

Overall, the suggested ensemble framework will be very useful in detecting misinformation in the age of fast-growing generative AI systems. This work can make a significant contribution to the overall cause of protecting the information ecosystems on the internet, with further extensions to multimodal, multilingual and adversarially-robust modelling.

References

- [1] Koustav Rudra, Niloy Ganguly, Jeanne Mifsud Bonnici, Eric Müller-Budack and Ritumbra Manuvie.: Disinformation and Misinformation in the Age of Generative AI. In: Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining (WSDM '25), Association for Computing Machinery, New York, USA (2025).
- [2] Su, Q., Wan, M., Liu, X. and Huang, C. R.: Motivations, methods and metrics of misinformation detection: an NLP perspective. In: Natural Language Processing Research, Vol. 1(1), pp. 1-13 (2020).
- [3] Abdali, S. et al.: Multi-modal misinformation detection: Approaches, challenges and opportunities. In: ACM Computing Surveys, Vol. 57(3), pp. 1-29 (2024).
- [4] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W. and Liu, T. Y.: LightGBM: A highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems (2017).
- [5] Essa, Ehab, Omar, et al.: Fake news detection based on a hybrid BERT and LightGBM models. In: Complex & Intelligent Systems, Vol. 9, pp. 6581-6592 (2023).
- [6] Guo, B., Ding, Y., Yao, L., Liang, Y. and Yu, Z.: The future of misinformation detection: new perspectives and trends. arXiv preprint arXiv:1909.03654 (2019).
- [7] Victoria L. Rubin and Tatiana Lukoianova.: Truth and deception at the rhetorical structure level. In: Journal of the Association for Information Science and Technology, Vol. 66(5), pp. 905-917 (2015).
- [8] Carlos Castillo, M. Mendoza and B. Poblete.: Information credibility on twitter. In: Proceedings of the 20th International Conference on World Wide Web, ACM, pp. 675-684 (2011).
- [9] Carlos Castillo, M. Mendoza and B. Poblete.: Predicting information credibility in time-sensitive social media. In: Internet Research, Vol. 23(5), pp. 560-588 (2013).

- [10] Qazvinian, V., Rosengren, E., Radev, D. and Mei, Q.: Rumor has it: Identifying misinformation in microblogs. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, pp. 1589-1599 (2011).
- [11] Martin Potthast, Johannes Kiesel, Kevin Reinartz, Janek Bevendorff and Benno Stein.: A stylometric inquiry into hyperpartisan and fake news. arXiv preprint arXiv:1702.05638 (2017).
- [12] Ke Wu, Song Yang and Kenny Q. Zhu.: False rumors detection on Sina Weibo by propagation structures. In: Proceedings of the IEEE 31st International Conference on Data Engineering, pp. 651–662 (2015).
- [13] Benjamin D. Horne and Sibel Adali.: Fake news packs a lot in title, uses simpler, repetitive content in text body, more similar to satire than real news. In: Proceedings of the Eleventh International AAAI Conference on Web and Social Media (2017).
- [14] Marco L. Della Vedova, Eugenio Tacchini, Stefano Moret, Gabriele Ballarin, Massimo DiPierro and Luca de Alfaro.: Automatic online fake news detection combining content and social signals. In: Proceedings of the IEEE 22nd Conference of Open Innovations Association (FRUCT), pp. 272-279 (2018).
- [15] Kai Shu, Suhang Wang and Huan Liu.: Exploiting tri-relationship for fake news detection. arXiv preprint arXiv:1712.07709 (2017).
- [16] Jing Ma, Wei Gao, Prasenjit Mitra, Sejeong Kwon, Bernard J. Jansen, Kam-Fai Wong and Meeyoung Cha.: Detecting rumors from microblogs with recurrent neural networks. In: Proceedings of the 25th International Joint Conference on Artificial Intelligence (IJCAI), pp. 3818-3824 (2016).
- [17] Lizhao Li, Guoyong Cai and Nannan Chen.: A rumor events detection method based on deep bidirectional GRU neural network. In: Proceedings of the IEEE 3rd International Conference on Image, Vision and Computing (ICIVC) (2018).
- [18] Yang Liu and Yi-Fang Brook Wu.: Early detection of fake news on social media through propagation path classification with recurrent and convolutional networks. In: Proceedings of the 32nd AAAI Conference on Artificial Intelligence (2018).
- [19] Han Guo, Juan Cao, Yazi Zhang, Junbo Guo and Jintao Li.: Rumor detection with hierarchical social attention network. In: Proceedings of the 27th ACM International Conference on Information and Knowledge Management, pp. 943–951 (2018).
- [20] Qi, P., Yan, Z., Hsu, W. and Lee, M. L.: Sniffer: Multimodal large language model for explainable out-of-context misinformation detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13052-13062 (2024).
- [21] Liu, Z., Zhang, X., Yang, K., Xie, Q., Huang, J. and Ananiadou, S.: FMDLlama: Financial misinformation detection based on large language models. In: Companion Proceedings of the ACM Web Conference, pp. 1153-1159 (2025).
- [22] Zhiwei Liu, Kailai Yang, Qianqian Xie, Christine de Kock and Eduard Hovy.: RAEmoLLM: Retrieval augmented LLMs for cross-domain misinformation detection using in-context learning based on emotional information. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, pp. 16508–16523 (2025).
- [23] Jiatong Ma, Linmei Hu, Rang Li and Wenbo Fu.: LoCal: Logical and causal fact-checking with LLM-based multi-agents. In: Proceedings of the ACM Web Conference 2025 (WWW '25), pp. 1614–1625 (2025).
- [24] Zikun Cui, Tianyi Huang, Chia-En Chiang and Cuiqianhe Du.: Toward verifiable misinformation detection: A multi-tool LLM agent framework. In: Proceedings of the 2025 International Conference on Generative Artificial Intelligence for Business (2025).
- [25] Soundankar, A.: Gen AI Misinformation Detection Data [Data set]. Kaggle. <https://www.kaggle.com/datasets/atharvasoundankar/gen-ai-misinformation-detection-dataset-20242025> (2024).
- [26] Sanh, V., Debut, L., Chaumond, J. and Wolf, T.: DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108 (2019).
- [27] Garreau, D. and von Luxburg, U.: Explaining the explainer: A first theoretical analysis of LIME. In: International Conference on Artificial Intelligence and Statistics (PMLR) (2020).
- [28] Mosca, E., Szigeti, F., Tragianni, S., Gallagher, D. and Groh, G.: SHAP-based explanation methods: a review for NLP interpretability. In: Proceedings of the 29th International Conference on Computational Linguistics, pp. 4593-4603 (2022).

© 2026 The Author(s). Published by WhiteField Academic Press. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. Authors retain copyright.



INTERNATIONAL JOURNAL OF ADVANCED COMPUTING AND MACHINE INTELLIGENCE

*Peer-Reviewed · Open Access · Published by WhiteField
Academic Press
Noida, India*



Scan to visit

www.ijacmi.org
editor@ijacmi.org
submissions@ijacmi.org

Visit our website — www.ijacmi.org

